

CSE 121 Project

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Authored by: Your Name

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Acknowledgement

We would like to thank Dr. Mohamed Omar and our TA Osama Tayel for the amazing work and effort that they showed us. With their help, guidance, and teaching we were able to enjoy this subject. We would also like to thank the faculty for all the resources and facilities that they provide to help us work efficiently. Finally, our families deserve the biggest thanks for their constant support in everything we do.
Thank you all!

Introduction

In this project we aim to build a calculator that can add, subtract and multiply 4 signed bits binary numbers together and display the result on seven segment displays. We did so by using ICs of various kinds which helped us build the circuit to do such functions.

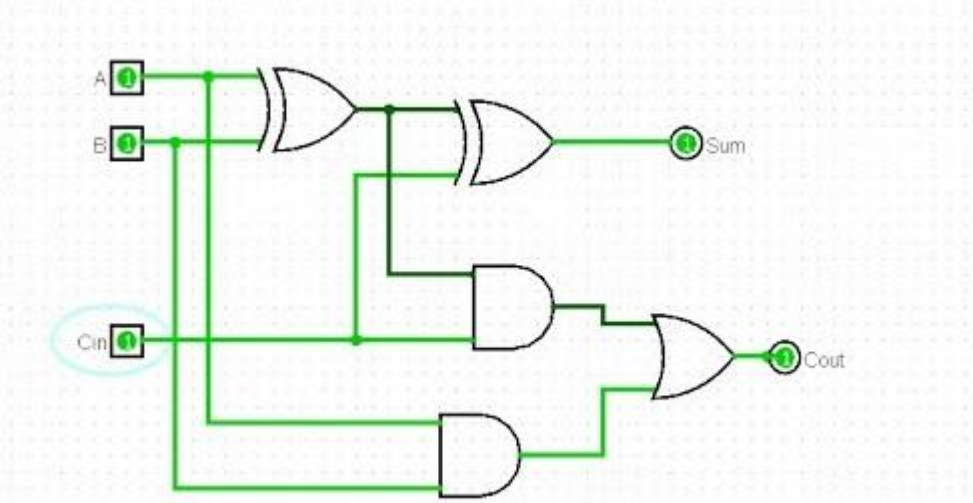
We will explain in full details how each function works to receive the desired output.

Methods

Addition:

Let us talk about the first function, the addition and how we were able to add 4-bit binary numbers together.

First, we must enter our inputs, let us call them A and B. A has 4 bits that we must input and same for B. The fourth bit is the signed bit, and its value will be 0 in the addition function. For each bit added from A and B a full bit adder will be used so 4 full bit adders are used to add each bit of the 4. The output is 4 bits, this is due to the carry which we added a separate full adder to it. If the final output is positive the final carry would be 0. Let us talk more about the full adders that were added and how they were made.



This is the full bit adder. Its aim is to output the sum of 2 bits and the carry of the sum if there is one.

The first bit and the second bit are inputted in an XOR gate also it is inputted in an AND gate. The output from the XOR gate is then entered through another XOR gate with the initial carry and that would be the final sum. Now to get the carry the output from the first XOR gate is also entered

through an AND gate with the carry. Finally, the output of the first AND and the second AND gates enters an OR gate.

Let's take an example to make it clearer.

1 bit plus 1 bit plus an initial carry of 1 is equal to 1 and carry 1.

So, 1 and 1 will enter the first XOR gate giving us a 0, they will also go through an AND gate giving us 1. Now the 0 from the XOR gate will enter another XOR gate with the initial carry which is 1 outputting 1 which is the final sum. The same 0 will also enter an AND gate with the initial carry which will give us 0. Now this 0 and the 1 from the first AND gate will enter an OR gate which will finally results in 1 which is the carry.

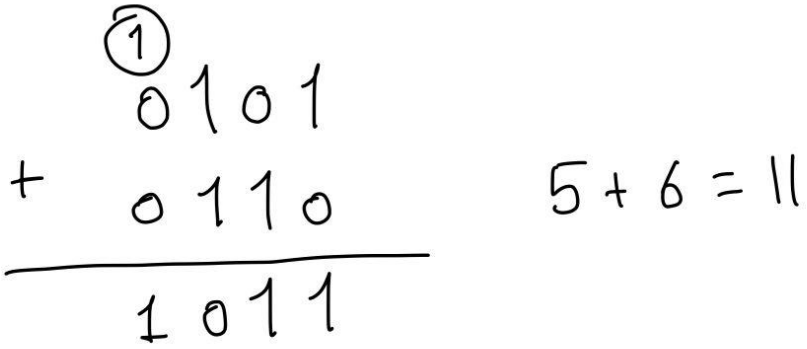
Now let's return to the addition function.

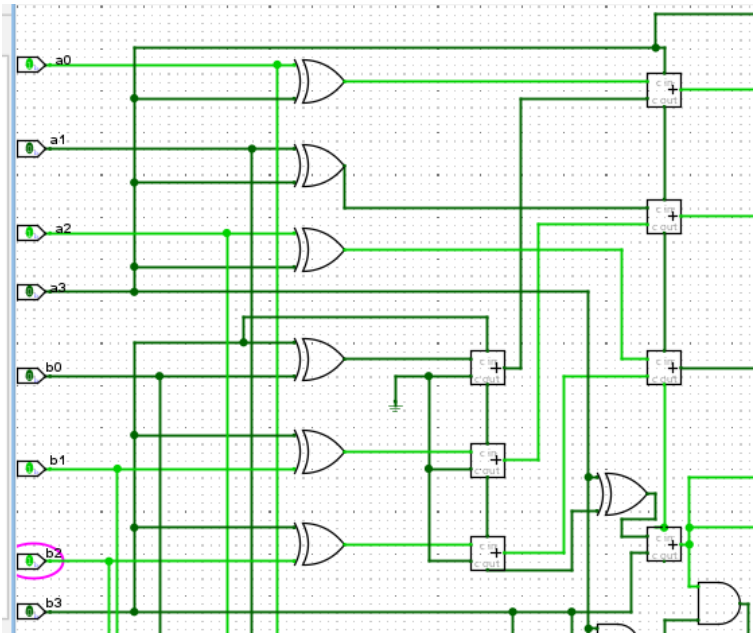
To explain it better let us use an example and see how each bit goes through the addition process.

We will use.

5+6=11

0101+0110=1011





Let’s add bit by bit and see how they travel through the circuit.

A0 and B0 (1+0 =1 with no carry)

The a0 enters an XOR with the a3 which is 0 because this is a positive

number. The same happens for b0 and b3. The output of the first XOR gate, which is 1 goes to a full bit adder with the initial being the a3. The output of the second XOR gate will go through another full bit adder with the ground value of 0 and the initial carry being b3. Now the output of this adder which is 0 will enter the first adder outputting 1.

A1 and B1 (0+1 =1 with no carry)

The a1 enters an XOR with the a3. The same happens for b1 and b3. The output of the first XOR gate which is 0 goes to a full bit adder with the initial carry being 0 which came from the previous adder. The output of the second XOR gate which is 1 will go through another full bit adder with the ground value of 0 with the initial carry being 0 which came from the previous adder. Now the output of this adder which is 1 will enter the first adder outputting 1.

A2 + B2 (1+1 =0 with carry 1)

The a2 enters an XOR with the a3. The same happens for b2 and b3. The output of the first XOR gate, which is 1, goes to a full bit adder with the initial carry being 0 which came from the previous adder. The output of the second XOR gate which is 1 will go through another full bit adder with the ground value of 0 with the initial carry being 0 which came from the previous adder. Now the output of this adder, which is 1 will enter the first adder outputting 0.

Now the final carry from the adder which has the ground and B2 XOR B3 will go an XOR gate with A3 and its output will go to a full bit adder with B3. This in turns gives us the final carry. (1011)

And the previous 3 adders gave us the correct values of the sum of each bit.

Subtraction:

To start the subtraction function we need to turn on the signed bit for each input A and B or either. Now that it is a negative number, we need to get 2 compliments for each A and B or either.

For A we pass it through an XOR gate and add 1 which is the initial carry in the full adder where addition occurs however for B, we pass it through XOR gate and then we use a different full adder to add 1 which is the initial carry.

We have 2 cases for the subtraction function.

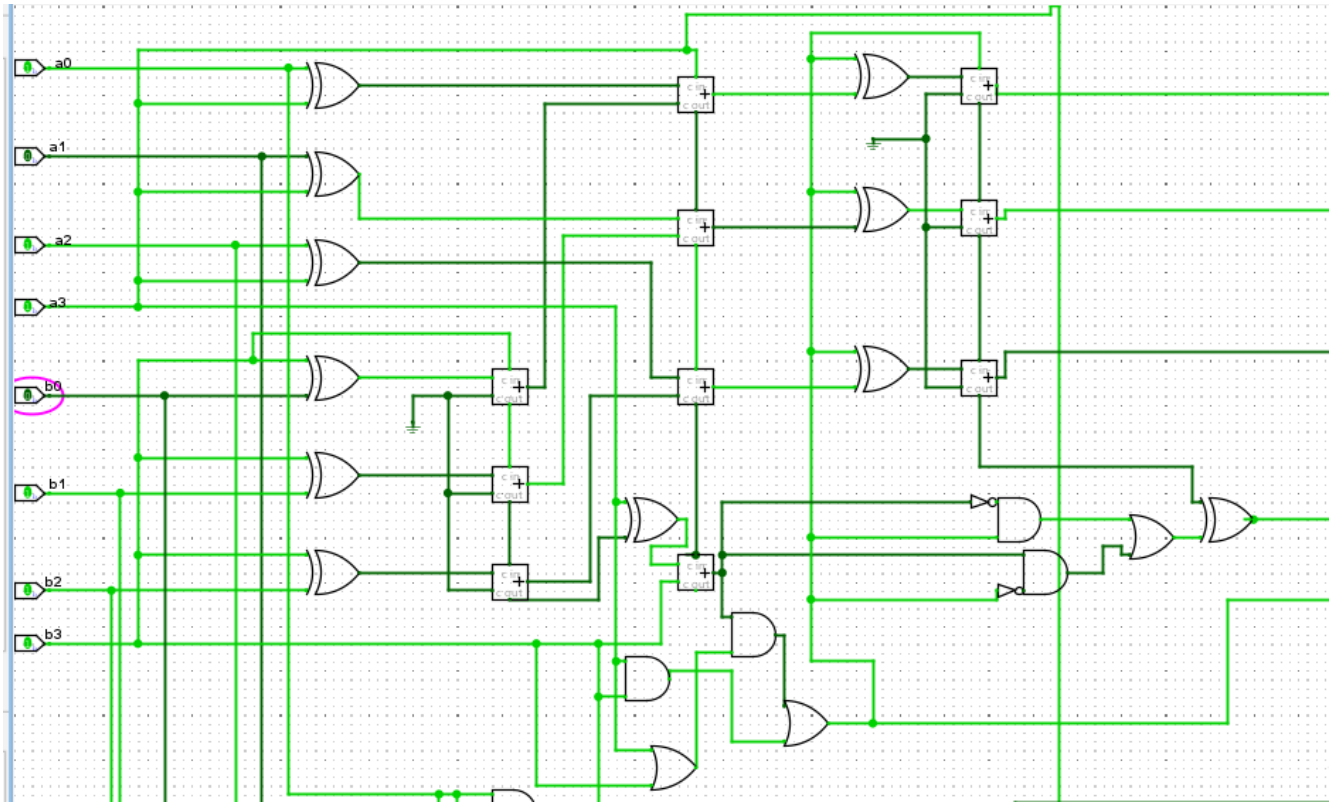
1) Positive – Negative
2) Negative – Negative

For the first case:

If the output is positive, we do not change it at all and leave it as it is. If not, we must get 2's compliment to the answer knowing the final bit is the signed bit.

For the second case:

The answer must change using 2's compliment because in all ways the answer would be negative.



Let us take bit by bit.

We will use an example to help us understand what is happening here.

(-5)+(-6)= (-11)

1101 + 1110 = 11011

-5 → 1101 (-5)+(-6)=(-11)

-6 → 1110

2's comp. of -5 → 0011

2's comp. of -6 → 0010

$$\begin{array}{r} 0\,\overset{1}{0}\,1\,1 \\ +\;0\,0\,1\,0 \\ \hline 0\;1\;0\;1 \end{array}$$

2's comp. of 0101 → 1011

-1011

A0+B0 (1+0=1 with no carry)

The a0 enters an XOR with the a3 which is 1 because this is a negative

number. The same happens for b0 and b3. The output of the first XOR gate, which is 0 goes to a full bit adder with the initial carry being the a3. The output of the second XOR gate, which is 1, will go through another full bit adder with the ground value of 0 and the initial carry being b3. Now the output of this adder which is 0 will enter the first adder outputting 1.

A1+B1 (0+1=1 with no carry)

The a1 enters an XOR with the a3 which is 1. The same happens for b1 and b3. The output of the first XOR gate, which is 1, goes to a full bit adder with the initial carry coming from the previous adder of 0. The output of the second XOR gate, which is 0 will go through another full bit adder with the ground value of 0 and the initial carry coming from the previous adder of 1. Now the output of this adder, which is 1 will enter the first adder outputting 0.

A2+B2 (1+1=0 with a carry of 1)

The a2 enters an XOR with the a3 which is 1. The same happens for b2 and b3. The output of the first XOR gate, which is 0, goes to a full bit adder with the initial carry coming from the previous adder of 1. The output of the second XOR gate, which is 0 will go through another full bit adder with the ground value of 0 and the initial carry coming from the previous adder of 0. Now the output of this adder which is 0 will enter the first adder outputting 1.

Now the final carry from the adder which has the ground and B2 XOR B3 will go to an XOR gate with A3, and its output will go to another full bit adder with B3 with its initial carry being 0 coming from the adder which has a2 XOR a3 and the sum from the adder which has the ground and B2 XOR B3

Now we have 4 full bit adders we will call them D1,D2,D3 and D4

We will take an extra step to ensure getting the correct answer.

We will connect a3 and b3 through an AND gate and an OR gate. The output from the OR gate will enter an AND gate with the sum from D4. The output of the first and second AND gates will go through an OR gate to ensure that the sign of the answer exists separately. Let's name the output K.

D1 will enter an XOR gate with K and its output will enter another adder lets call it E1 with the ground value of 0 and its initial carry being K outputting 1.

D2 will enter an XOR gate with K and its output will enter another adder lets call it E2 with the ground value of 0 and its initial carry coming from E1 and its value is 0 outputting 1.

D3 will enter an XOR gate with K and its output will enter another adder lets call it E3 with the ground value of 0 and its initial carry coming from E2 and its value is 0 outputting 0.

To also make sure that the sign is in a different wire and not in the same wire as the magnitude we will get the sum of D4 and input it through a NOT gate then an AND gate with K, also use D4 and input it in another AND gate with NOT K. The output of both AND gates will go through an OR gate and the output will go through an XOR gate with the carry from E3 outputting 1.

So, the final result would be :

11011 with the first 1 on the left being the signed bit which will appear on a separate seven segment.

Multiplication

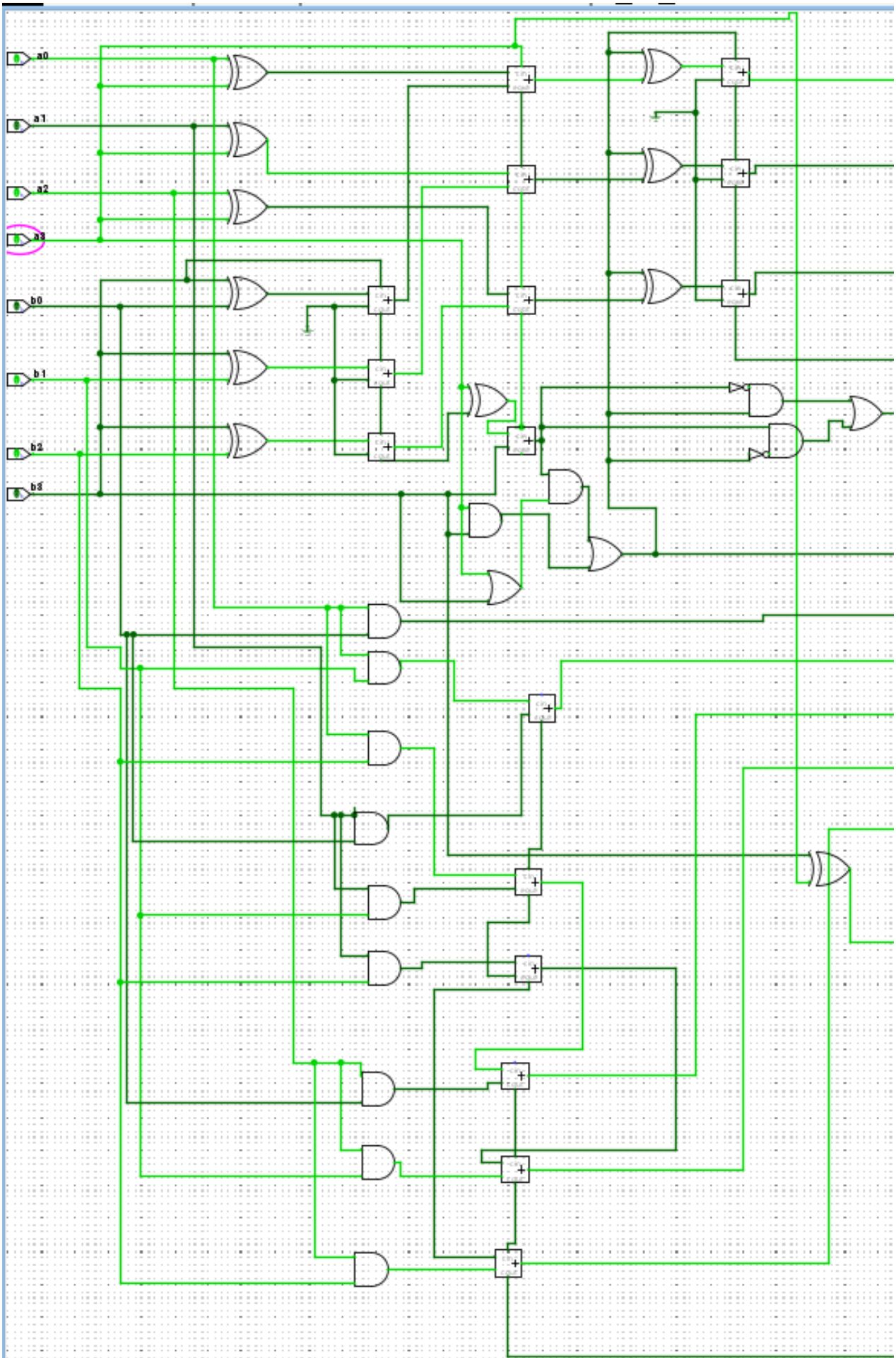
To start the multiplication process we need to turn on the switch in the circuit. If its value is 1 multiplication occurs which is what we currently need.

We have 3 cases for the multiplication function.

1) Positive * positive = positive
2) Positive * negative = negative
3) Negative * negative = positive

To solve the negative an XOR gate would be used but we will know more with circuit.

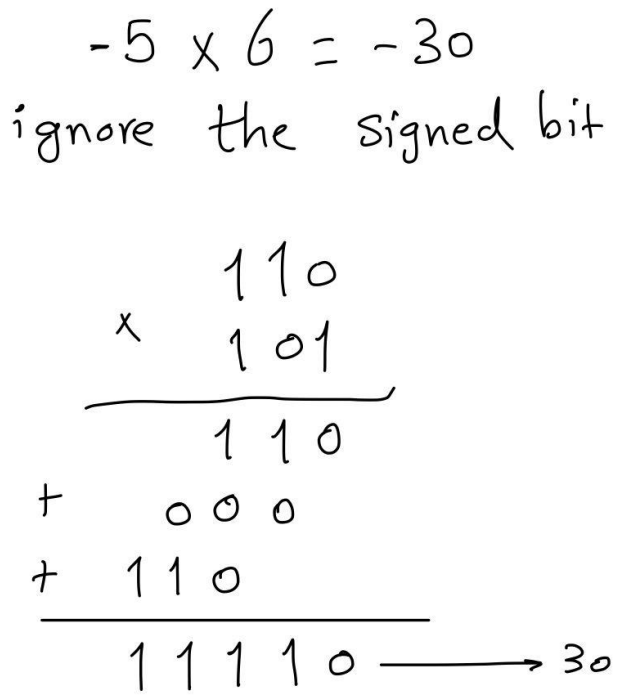
In the breadboard we did not use directly the full adder but used its gates.


a0
a1
a2
a3
b0
b1
b2
b3

Let us explain the circle better with an example.

We will use.

(-5) * 6 = -30



We will ignore the sign , consequently we will ignore the signed bit, but will add it to the final answer later.

So, what we will be working is

101*110=11110

In multiplication we need to multiply each term from A for example with all the terms of B.

A0*B0 and A0*B1 and A0*B2

(1*0 and 1*1 and 1*1)

Both a0 and b0 will enter an AND gate and the output 0 would be considered the first bit of the final answer.

Both a0 and b1 will enter an AND gate and the output 1 will enter full bit adder which we will call F1 with no initial carry.

Both a0 and b2 will enter an AND gate and the output 1 will enter full bit adder which we will call F2 with an initial carry of 0 coming from F1

A1*B0 and A1*B1 and A1*B2

(0*0 and 0*1 and 0*1)

Both a1 and b0 will enter an AND gate and the output 0 will enter F1.

F1 will now output the second bit of the final answer.

Both a1 and b1 will enter an AND gate and the output 0 will enter F2 and the output 1 will enter another adder which we will call F4. F2's final carry will be inputted in another adder which we will call F3.

Both a1 and b2 will enter an AND gate and the output 0 will enter F3. F3 has no initial carry. F3's output 1 will enter another adder which we will call F5.

F3's final carry will be inputted in another adder which we will call F6.

A2*B0 and A2*B1 and A2*B2

(1*0 and 1*1 and 1*1)

Both a2 and b0 will enter an AND gate and the output 0 will enter F4. Now the output of F4 will be the third bit of the final answer.

Both a2 and b1 will enter an AND gate and the output 1 will enter F5.

F5 will have an initial carry 0 coming from F4. Now the output of F5 will be the fourth bit of the final answer.

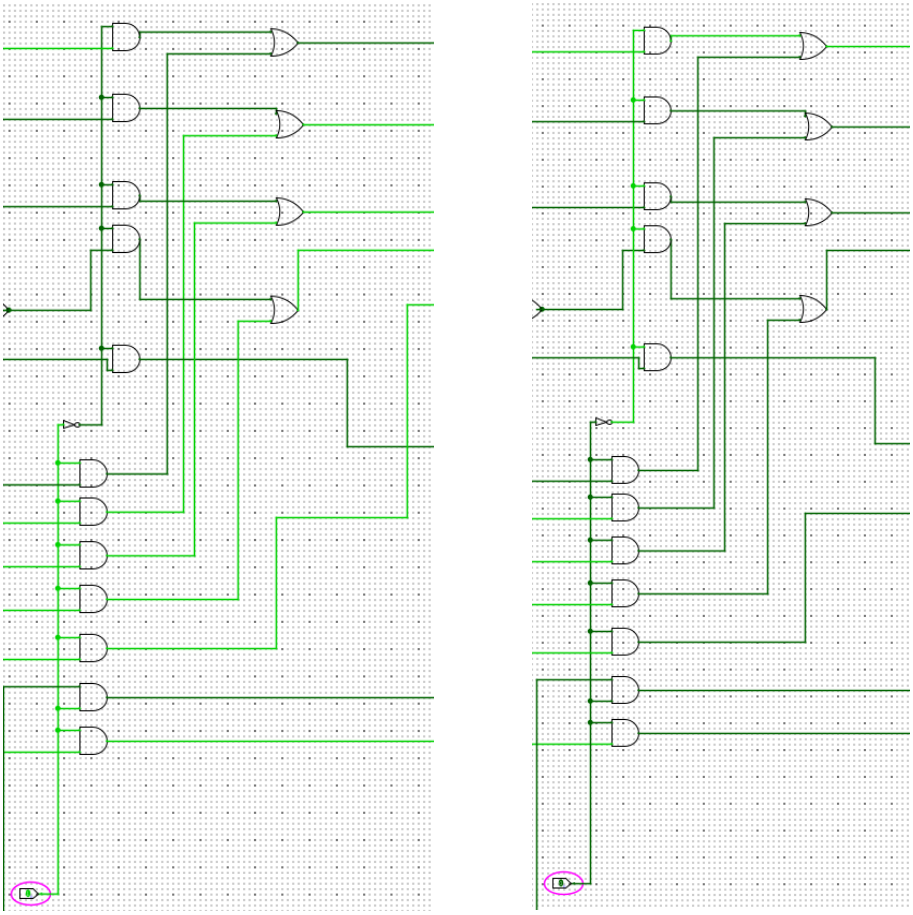
Both a2 and b2 will enter an AND gate and the output 1 will enter F6.

F6 will have an initial carry 0 coming from F5. Now the output of F6 will be the fifth bit of the final answer.

Now we have our final answer 11110, but it is missing its sign.

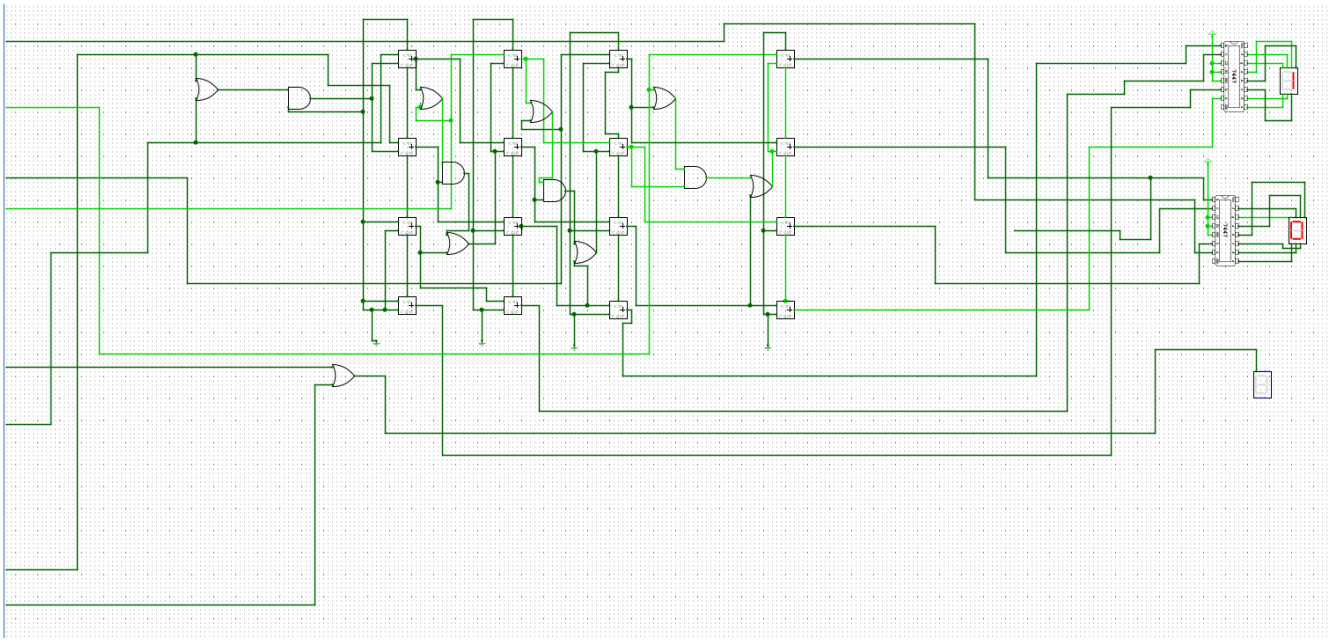
As we know we assigned the sign bit to a3 and b3. We will pass both bits through an XOR gate. As we know if we input 2 different values in an XOR gate we get 1 and if they are alike, we get 0. Now if no signed bits are turned on, we would get 0 same happens if both signed bits are turned on. In our case only 1 is turned on, so now when it passes through the XOR gate we will get 1 which is the negative sign that is needed to complete our answer.

Selection



Using this switch, we will be able to switch between addition or subtraction and multiplication. Both operations addition and multiplication occur but we control the output using the switch. When its value is 0 it goes through a NOT gate so now it will enter 5 AND gates. Each AND gate from the 5 receive the value from the NOT gate and each bit of the final answer from the addition with the fifth AND gate receiving the signed bit. Their outputs are then inputted into 4 OR gates. If the switch is 1 multiplication occurs. The 1 enters 7 AND gates with each bit of the final answer of the multiplication with the seventh AND gate receiving the signed bit. The first 4 AND gates are the second input to the OR gates.

Binary to BCD



Since we have 6 outputs that we need to switch from binary form to BCD so we can use the 7-segment display we will use the shift add 3 method which offers less circuitry and more suitable way to prepare the output to display

Let's start talking about the shift add3 method.

It consists of 3 steps shifting checking adding.

Shifting: we need to shift our inputs to the left

Checking: after shifting we check if our inputs are bigger than or equal to 5 and we keep shifting till we finish our inputs

Adding: after completing the checking phase, the output of the checking phase is either a 0 or 1 if it is a 0 nothing is needed to be added (0 is added) but if 1 is the output of the check, 3 is added to the shifted number

We repeat all those steps for each bit of the number. The output of this process will convert a 6-bit binary input into an 8 bit BCD output

How to implement this process in logic gates and circuits

Preparing the inputs

6 input binary number is made of 6 bits each bit is corresponding to a certain value.

1 2 4 8 16 32

the 1 value is directly headed to the 7 segments part since this value does not affect the next part of the circuitry.

all 2 4 8 values are the inputs for 3 separate full adders IC.

16 32 values are the inputs for the 4th full adder IC.

The a3 a2 values for each full adder IC is connected to the ground while the a1 a0 is connected together preparing the (0011) inputs or 3 value for the add 3 part

To implement the shifting part in logic gates the sum output of each bit is connected as the next bit input (s0 is connected to b1,s1 is connected to b2,s2 is connected to b3)

The s3 output of 4th 3rd 2nd full adders are directly headed to the 7-segment part.

Checking part

While s0 is connected to b1 of the following full adder, it is also connected as an input with b0 of the following input using OR gate and the output is an input for AND gate and the 2nd input is s1 and the output is finally connected to a OR gate and its 2nd input is s2 ,the output is connected to the add3 part

so, when s2 equals 1 we add 3 or when s1 equals 1 and b0 of the next full adder is on (101 or 5 so we add 3) finishing the checking part and powering the add3 part.

As mentioned s3 of each full adder is headed to the 7 segments part but for the 1st full adder which is connected to the 2 value all s0 s1 s2 s3 is headed to the 7-segment part.

The 1 value is the LSB.

form the 1st full adder.

s0 is the 2nd bit.

s1 is the 3rd bit.

s2 is the 4th bit.

finishing the first 4 BCD bits

s3 is the LSB.

s3 of 2nd full adder is 2nd bit.

s3 of 3rd full adder is 3rd bit.

s3 of 4th full adder is 4th bit.

finishing the other 4 BCD bits converting 6 inputs binary to 8 BCD outputs

those 8 bits are now made of 2 groups each 4 is connected to a 7 segments display

we use a decoder to

1 convert form binary to BCD but it is useless in this case since it is already converted.

2 take the inputs and power the 7 segments display.

The decoder (7447 and 7448)

7447 is used for common anode 7 segment display while 7448 used for common cathode 7 segments displays.

Both ICs have 4 inputs 7 outputs those 7 outputs (a,b,c,d,e,f,g) will be connected to the 7 segment display inputs and based on the type of the 7 segment display (common anode, common cathode) the power will be connected to positive/ negative terminal for the display

For our case, the IC will have 4 inputs the first BCD outputs from the previous circuit and will light the first 7 segment display which will corresponds for the units of the decimal output of the whole project.

The second 4 inputs will be connected to the second decoder and power the second 7 segments display which displays tens of decimal output of the whole project.

Also if power was given to one of the inputs of the display itself a single led from the 7 will light up and that will be used to indicate if the number is positive or negative so if it is negative the g led in the 7 segment will be lit making the – sign appear

Conclusion

We were able to successfully build the three functions and output them correctly on the 7 segment displays. Tinker CAD, Logisim and the physical breadboard all worked as expected. In the addition we add bit by bit as for the subtraction we also add but any negative number must be changed to 2s complement and as discussed before the final output varies depending on the case provided. With multiplication we tried to implement how any multiplication is solved on paper and design it in the circuit. The switch provides an interesting approach on choosing a different mode either addition and subtraction or multiplication. Finally, all the outputs which were in binary had to be changed to BCD to be able to be used in the seven segment displays. This was achieved by the shift add 3 methods. We used this method as we had 6 outputs that had to be converted to BCD hence, we did not use the traditional method of adding 6.

References

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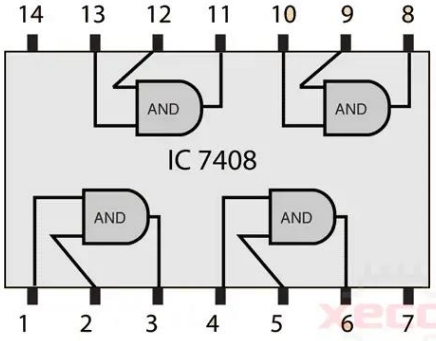
[4] "Combinational Circuits Part 1," Lecture Notes, Faculty of Engineering, Ain Shams University. [Online]. Available: https://lms.eng.asu.edu.eg/pluginfile.php/609369/mod_resource/content/1/07%20-%20Combinational%20Circuits_1.pdf.

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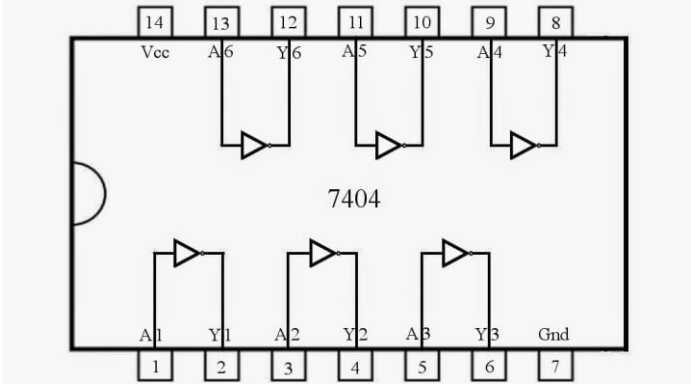
[6] Neso Academy, "Simplification using Boolean Algebra (Part 1)," YouTube, Sep. 11, 2017. [Online]. Available: https://youtu.be/sdH2lKiuCjg?si=Ibz4QwfQygYx8SYF.

Appendix

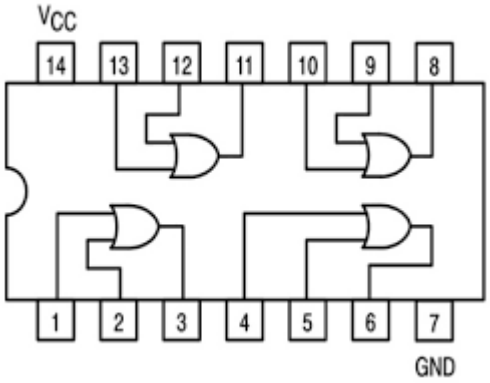
AND IC



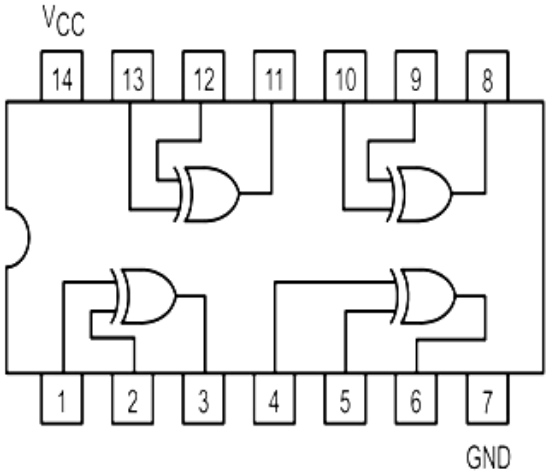
NOT IC



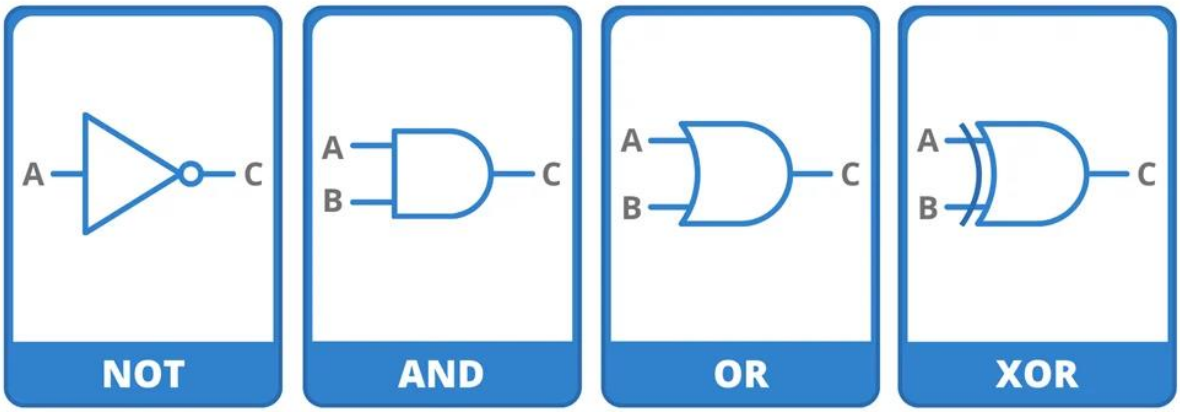
OR IC



XOR IC



AND, OR, NOT, XOR truth tables



A	C
0	**1**
1	**0**

NOT

A	B	C
0	0	**0**
0	1	**0**
1	0	**0**
1	1	**1**

AND

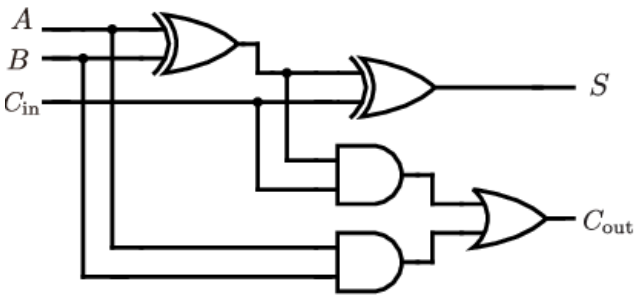
A	B	C
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**1**

OR

A	B	C
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**0**

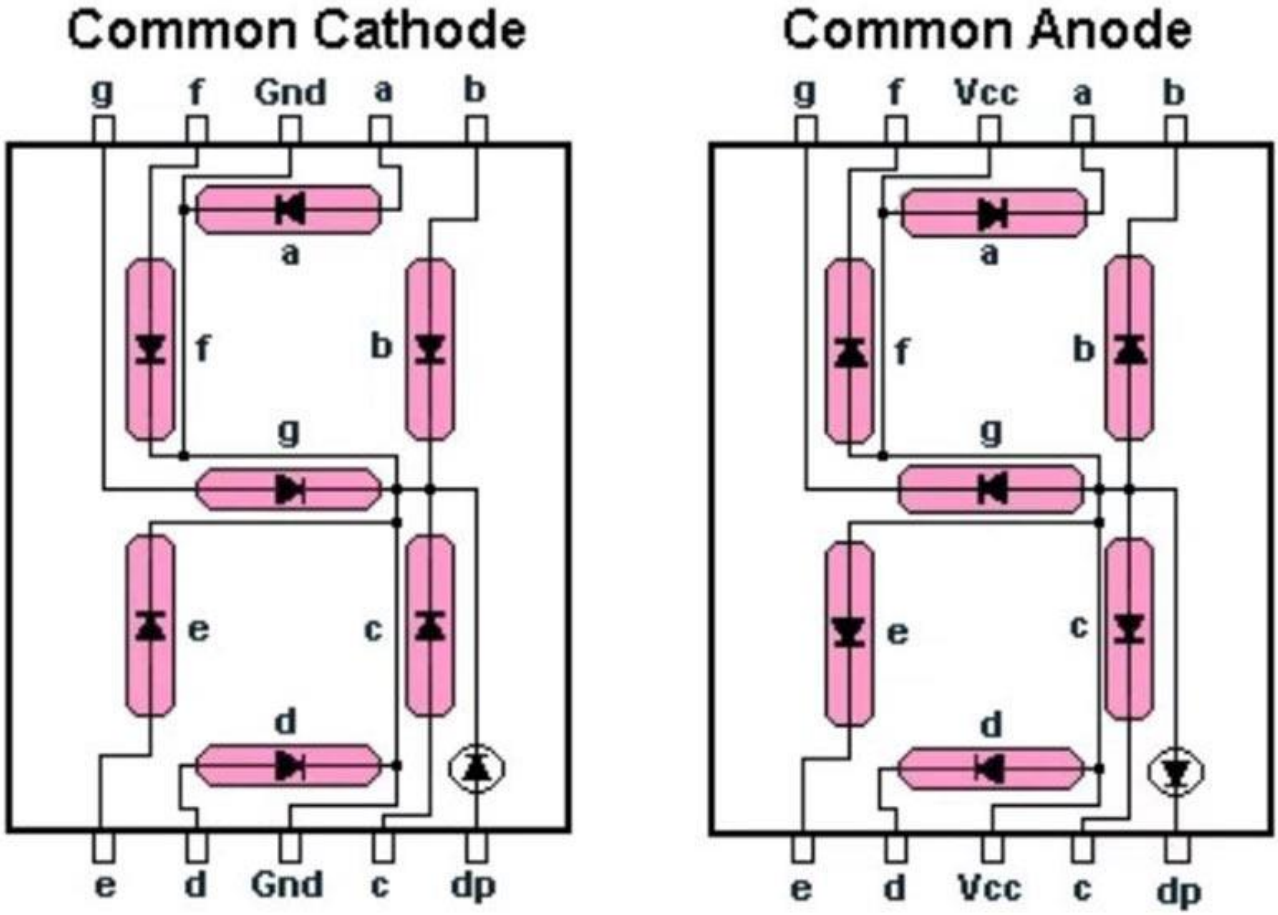
XOR

Full adder IC and truth table



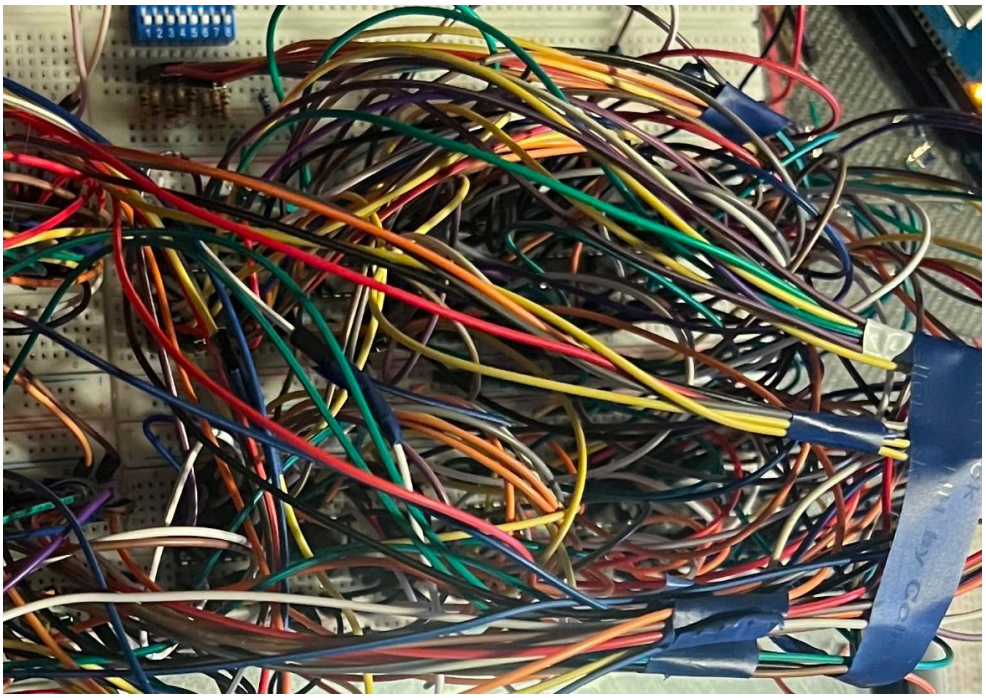
Inputs			Outputs	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Common cathode and common anode seven segment display pinout

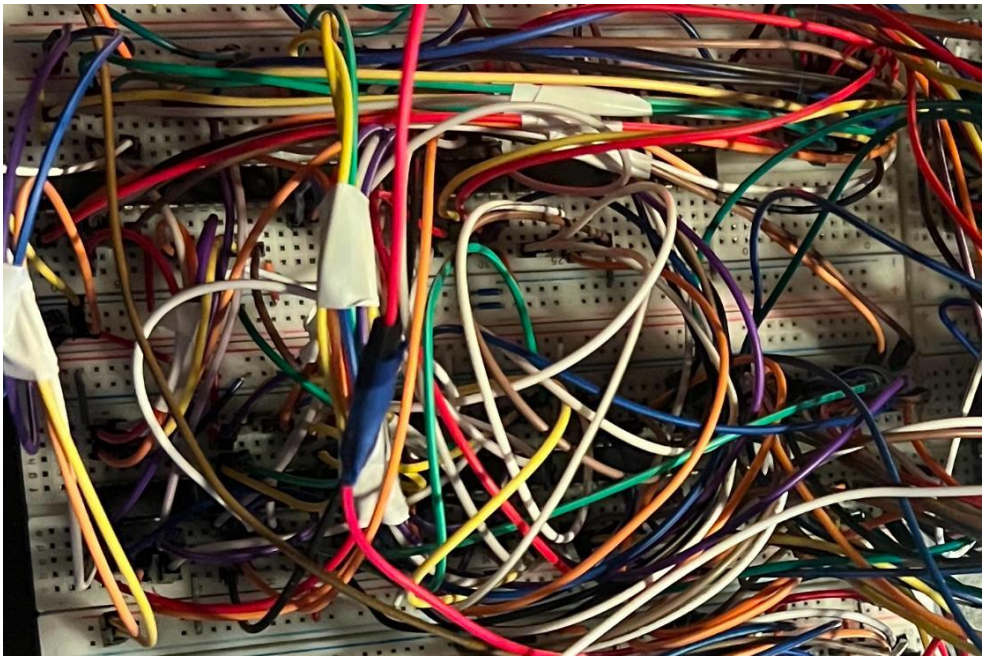


SSD Configuration

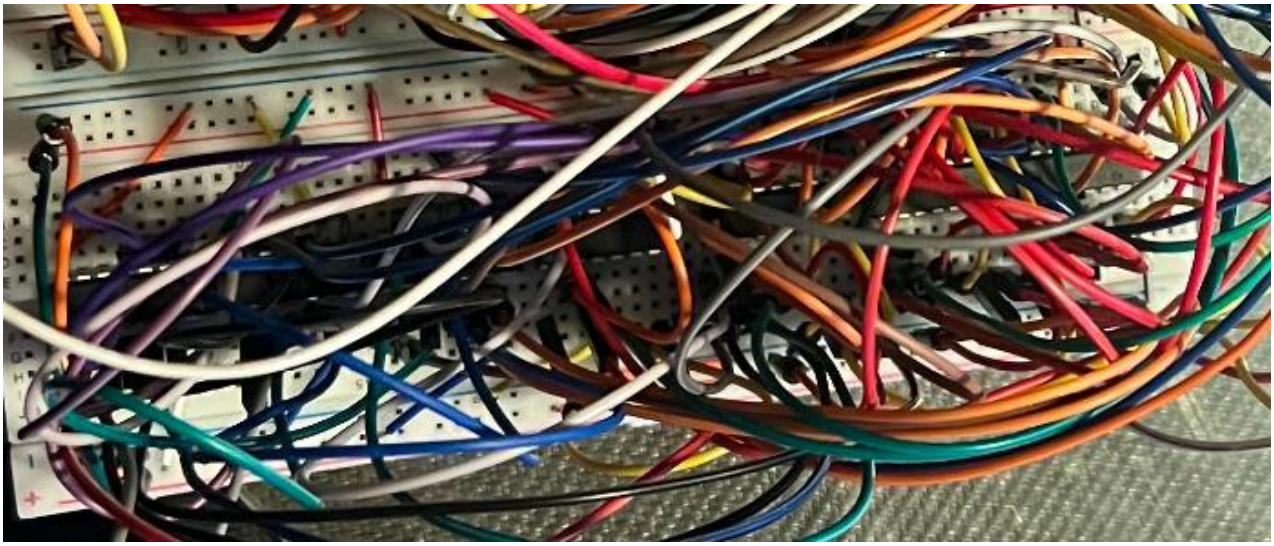
Addition and selection (physical)



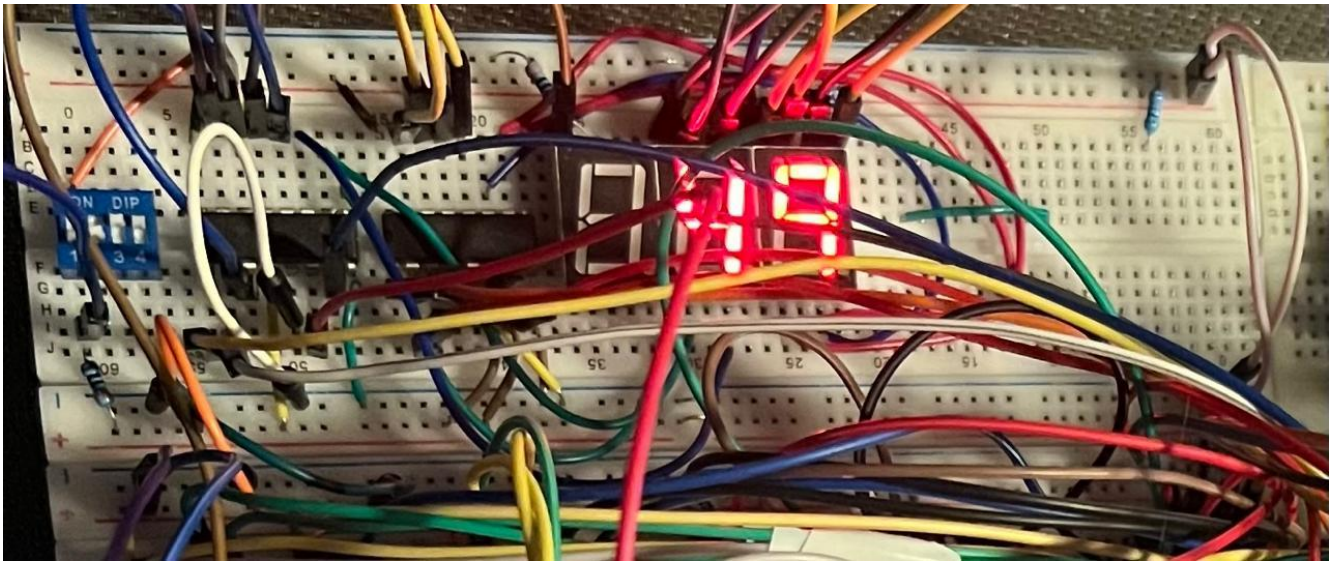
Multiplier (physical)



Binary to BCD (physical)



Seven segment display (physical)



Full circuit on Logisim

